

Project 3: SQL Data Analysis Report

Submission for DecodeLabs Internship Portal

1. Database & Table Setup

To analyze the required data patterns, a relational schema for tracking sales across regions was constructed using SQLite. The `sales_records` structure tracks order codes, customers, multi-category inventory metrics, transactional values, and geographic locations.

```
CREATE TABLE sales_records (  
    order_id INT,  
    customer_name VARCHAR(50),  
    product_category VARCHAR(50),  
    amount INT,  
    city VARCHAR(50)  
);  
  
INSERT INTO sales_records VALUES  
(101, 'Rahul', 'Electronics', 15000, 'Chennai'),  
(102, 'Priya', 'Clothing', 2500, 'Bangalore'),  
(103, 'Amit', 'Electronics', 45000, 'Mumbai'),  
(104, 'Sneha', 'Books', 1200, 'Chennai'),  
(105, 'Vikram', 'Clothing', 3500, 'Chennai'),  
(106, 'Rohan', 'Electronics', 12000, 'Bangalore'),  
(107, 'Anjali', 'Books', 800, 'Mumbai');
```

2. Analytical Query 1: Location Filter (WHERE, SUM, AVG)

Objective: Aggregating structural revenue performance data filtered specifically for the Chennai hub location.

```
SELECT city,  
       SUM(amount) AS total_revenue,  
       AVG(amount) AS average_spend  
FROM sales_records  
WHERE city = 'Chennai';
```

Result Output Set:

city	total_revenue	average_spend
Chennai	19700	6566.67

3. Analytical Query 2: Structural Aggregations (GROUP BY, COUNT)

Objective: Segregating overall operational volume profiles grouped metrics by categorized product streams.

```
SELECT product_category,
       COUNT(order_id) AS total_orders,
       SUM(amount) AS category_revenue
FROM sales_records
GROUP BY product_category;
```

Result Output Set:

product_category	total_orders	category_revenue
Books	2	2000
Clothing	2	6000
Electronics	3	72000

4. Analytical Query 3: Positional Sorting (ORDER BY)

Objective: High-to-low financial tracking prioritization sequence evaluating highest baseline transactions.

```
SELECT customer_name, product_category, amount
FROM sales_records
ORDER BY amount DESC;
```

Result Output Set:

customer_name	product_category	amount
Amit	Electronics	45000
Rahul	Electronics	15000

customer_name	product_category	amount
Rohan	Electronics	12000
Vikram	Clothing	3500
Priya	Clothing	2500
Sneha	Books	1200
Anjali	Books	800